

Project IV

Part I – Basic Probability

A die is thrown 3 times. Determine the probability of the event A that the resulting sequence of three integers is strictly increasing.

Parti II – Conditional Probability

A gambler throws the dice 1 and 2 simultaneously. What is the probability that die 1 shows a 6 (event A) on condition that both dice showed an even number (event B).

Part III – Bayes' Theorem (CHALLENGING)

Suggestion: Watch this [YouTube video](#) first.

1% of the population in a country are HIV-positive. A test procedure for diagnosing whether a person is HIV-positive indicates with probability 0.98 that the person is HIV-positive if indeed he/she is HIV-positive, and with probability 0.96 that this person is not HIV-positive if he/she is not HIV-positive.

- 1) What is the probability that a test person is HIV-positive if the test indicates that? (Hint: $P(B|A)=?$)
- 2) The probability that a person is HIV-negative if the test procedure indicates that? (Hint: $P(\bar{B}|\bar{A})=?$)

To solve the problem, random events A and B are introduced:

A = 'The test indicates that a person is HIV-positive.'

B = 'A test person is HIV-positive.'

From the problem:

$$P(B) = 0.01, P(A|B) = 0.98, P(\bar{A}|\bar{B}) = 0.96$$

Hints: Construct a tree diagram first. Then compute $P(A) = P(A|B) P(B) + P(A|\bar{B}) P(\bar{B})$ (total probability rule)